# Leveraging LLMs for automated reward function modelling in RL

Johan Lindqvist, Lennart Wittke
University of Washington

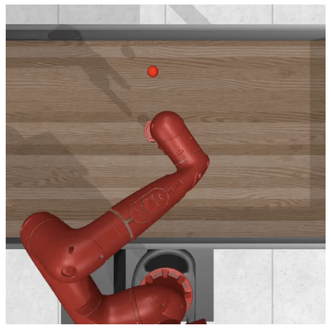
## Problem Statement and Research Question

Research indicates that the performance of RL algorithms heavily depends on the form of the reward function. Yet, designing an effective reward function is not trivial and often requires tedious trial and error, as well as human understanding of the problem at hand.

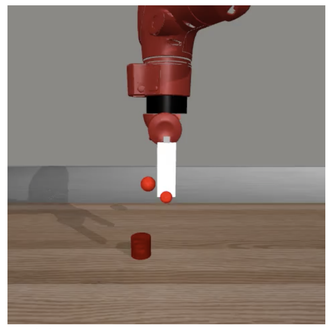
We explore is the possibility of using VLMs in order to better understand the environment that agents operate in and use that understanding to create automated rewards to incentivize the model to complete the overall objective.

**Research Question.** *Can VLMs (in a zero-shot fashion) serve as effective reward models by leveraging text prompts and environmental renderings?*

## Metaworld Environment and Camera Angles



(a) 'Topview' camera angle



(b) 'BehindGripper' camera angle

Figure: Robotic Setup Configurations

We selected MetaWorld for its simplicity and 3D image access, specifically using the 'reacher-v2' task, which is both easy and fast to train.

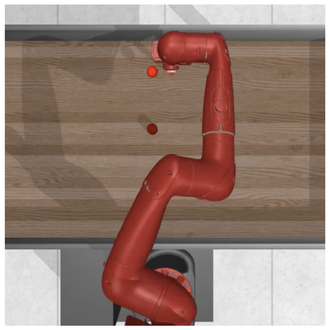
**The task's goal is to move the robot's end-effector as close as possible to the red ball target.**

We also explored multiple camera angles to determine their impact on learning. In particular, we decided to chose the top-view and behind-gripper perspectives, as we believe that their combination provides a comprehensive view of both the robot's vertical and longitudinal position in 3D space.

**Base RL Architecture:** We used a standard Soft-Actor-Critic (SAC) implementation as the base architecture, with $10$k seed steps, $100$k environment steps and a learning rate of $3e-4$. [2]

## Vision Language Models (VLMs) and Algorithms

**Approach 1.** We used the Molmo model [1], an LLM specifically designed to exhibit strong spatial understanding from images. Specifically, we propose the simple approach of providing an input image of the current state along with a prompt, and then asking the VLM to generate a reward value in response. Additionally, we prompted the model to reason through its decision, similar to the chain-of-thought prompting paradigm [4].



(a) Input image as state observation

**Molmo**

**Prompt:** *You are a reinforcement learning expert assisting with training an agent in the MetaWorld Reacher environment. Your task is to assign a single reward value based on the input observation, represented as an image. The image depicts [...] a number between 0 and 1 with an explanation for that value.*

**Response:** Based on the image, the robotic arm appears to be quite close to the target, with its end-effector positioned near the red ball. The arm's joints are well-aligned [...] is higher than 0.5 to acknowledge the arm's proximity to the goal and its generally good alignment.**Reward: 0.85**

(b) Molmo Output

Figure: Approach 1

**Approach 2.** CLIP is a model that was trained to ensure that the embeddings of images and text have a high cosine similarity if their semantics align [3]. Consequently, the core idea of this approach is to assign a high reward when the goal prompt and input image are closely aligned, indicating that the current state is near the goal.

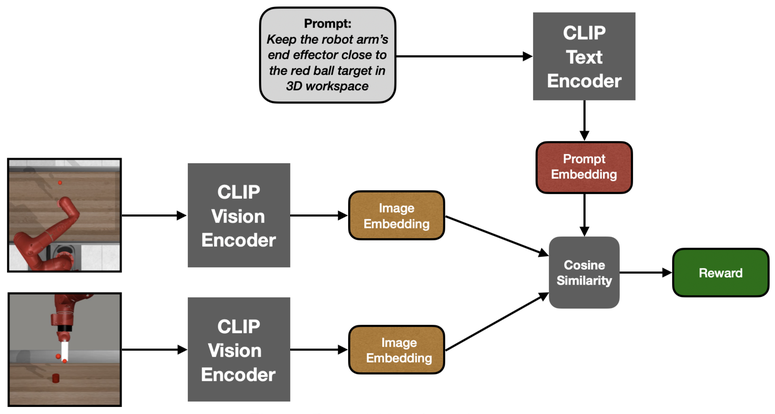



Figure: Reward Model architecture

## Results

| Reward Model | Average Episode Reward | Training Time (hrs) |
|---|---|---|
| Ground Truth Rewards | 4732.92 | 3.5 |
| Molmo (8B) | - | 100+ |
| CLIP-based approach (0.96B) | 437.43 | 8 |
| CLIP-based approach (1.37B) | 605.84 | 10 |
| Random Actions | 418.28 | 0 |

Table: Average Episode Reward after $100$k training steps. Trained on L4 GPU.

- We observe that the CLIP based model performs much worse than the ground truth baseline and only slightly outperforms random actions.
- The added inference time from the large VLM's forward passes significantly increases training time. With the first approach, training the model within a reasonable timeframe was not possible at all.

## Conclusion

- There are promising results where MOLMO demonstrates the ability to understand the problem and provide reward scores suitable for the tasks, given carefully engineered prompts.
- However, MOLMO struggles with scaling the rewards, even when given reference values, and can occasionally misinterpret events occurring across sequences of images.
- The CLIP-based approach has a much lower inference time, but it does not seem to represent the problem distinctly enough to compete with an engineered reward function.
- Possible improvements to Approach 2 should focus on improving the embeddings, especially to utilize models with better spatial understanding of 3D-environments.
- Future research could focus on improving inference efficiency and finding prompts that can be more effectively used to generate reward functions with VLMs.

## References

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[2] Tuomas Haarnoja, Aurick Zhou, Pieter Abbeel, and Sergey Levine. Soft actor-critic: Off-policy maximum entropy deep reinforcement learning with a stochastic actor. In *International conference on machine learning*, pages 1861–1870. PMLR, 2018.

[3] Alec Radford, Jong Wook Kim, Chris Hallacy, Aditya Ramesh, Gabriel Goh, Sandhini Agarwal, Girish Sastry, Amanda Askell, Pamela Mishkin, Jack Clark, et al. Learning transferable visual models from natural language supervision. In *International conference on machine learning*, pages 8748–8763. PMLR, 2021.

[4] Jason Wei, Xuezhi Wang, Dale Schuurmans, Maarten Bosma, Fei Xia, Ed Chi, Quoc V Le, Denny Zhou, et al. Chain-of-thought prompting elicits reasoning in large language models. *Advances in neural information processing systems*, 35:24824–24837, 2022.

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